

Professional Profile

I'm an enthusiastic and driven computational chemist with strong background in quantum mechanics, molecular dynamics, and data analysis. My work spans ion channel permeation, protein glycation, and high-throughput screening of metal-organic frameworks, supported by expertise in Python and packages like VASP, Gaussian, GROMACS, Rosetta, and AlphaFold, etc.

With graduate degrees in chemistry and chemical engineering, I'm passionate about using computational methods to solve complex problems in chemistry and biology. I thrive in interdisciplinary teams and enjoy mentoring, presenting, and contributing to impactful research.

Education

Virginia Polytechnic Institute and State University	M.S. Chemistry, 2021 – 2025
New Mexico State University	Ph.D. Chemical Engineering, 2016 – 2020
East China University of Science and Technology	B.E. Chemical Engineering, 2012 – 2016
New Mexico State University	B.S. Chemical Engineering, 2015 – 2016

Research Experience

Extensive experience in running large-scale quantum chemistry and molecular dynamics simulations on high-performance computing (HPC) systems and GPU clusters. Skilled in job scheduling with SLURM, automating simulation workflows with Python.

Graduate researcher in Dr. Valerie Welborn's group at Virginia Tech, Jan 2021 – May 2025

- Conducted extensive homology, conformational and molecular dynamics (MD) modeling of ion permeation in human voltage-gated sodium channel and glycation of tropoelastin.
- Implemented a Python pipeline to analyze the structure and dynamics of protein hydration water.
- Developed a Python-based analysis script for MD trajectories, achieving a 30% improvement in efficiency compared to VMD.

Graduate researcher in Dr. Thomas Manz's group at NMSU, Aug 2016 – Dec 2020

- Developed Python pipeline that screens thousands of metal-organic frameworks (MOFs) for defects.
- Established a database of computational ready MOFs with forcefield precursors.
- Created a robust and efficient method to calculate atomic polarizabilities and London dispersion coefficients from quantum chemistry simulations.
- Formulated a set of comprehensive principles to explain chemical bonding and electron redistribution between angular momentum subshells.

Publications

- Y. Zheng, **T. Chen**, and V. V. Welborn. "Preorganized electric fields in voltage-gated sodium channels", *ChemBioChem*, 2025.
- W. R. Archer, **T. Chen**, V. V. Welborn, and M. D. Schulz. "Polymer Tacticity Effects in Polymer–Lanthanide Chelation Thermodynamics", *Macromol*, 2023.
- **T. Chen**, T. A. Manz, "Identifying misbonded atoms in the 2019 CoRE metal–organic framework database", *RSC. Adv*, 2020.
- **T. Chen**, T. A. Manz, "A collection of forcefield precursors for metal–organic frameworks", *RSC. Adv*, 2019.
- **T. Chen**, T. A. Manz, "Bond orders of the Diatomic Molecules", *RSC. Adv*, 2019.
- T. A. Manz, **T. Chen**, D. J. Cole, N. Gabaldon Limas, and B. Fiszbein, "New scaling relations to compute atom-in-material polarizabilities and dispersion coefficients: part 1. Theory and accuracy", *RSC. Adv*, 2019.
- T. A. Manz, **T. Chen**, "New scaling relations to compute atom-in-material polarizabilities and dispersion coefficients: part 2. Linear-scaling computational algorithms and parallelization", *RSC. Adv*, 2019.

Skills

- Programming language: Python, with extensive experience in **SciPy**, **NumPy**, **pandas**, **Matplotlib**, and MDAnalysis.
- Simulation software: **VASP**, **Gaussian**, Quantum Espresso, **GROMACS**, Tinker.
- Visualization software: **VMD**, **PyMOL**, VESTA, GaussView.
- Other packages: **Rosetta**, **AlphaFold**, SWISS-MODEL, Modeller, CCDC database and software.
- Language: Chinese (native), English (bilingual).

Leadership Activities

Chemical Engineering Graduate Student Organization Treasurer at NMSU, 2017 – 2018

Coordinated numerous organizational activities in collaboration with fellow officers, including meetings, volunteer initiatives, and fundraising events.

Presentations

- **T. Chen**, Y. Zheng, M. Lawal, and V. Welborn, “Identifying key residues in permeation process through ion channels with electric field”, ACS Meeting Spring 2024, New Orleans, Louisiana, March 20, 2024.
- **T. Chen**, Y. Zheng, and V. Welborn, “Extending Electrostatic Preorganization Theory to Ion Channels”, Southeastern Theoretical Chemistry Association Meeting, Blacksburg, Virginia, May 16, 2024.
- **T. Chen**, Y. Zheng, M. Lawal, and V. Welborn, “Atomistic Modeling of Voltage-Gated Sodium Ion Channel”, Southeastern Theoretical Chemistry Association Meeting, Atlanta, Georgia, May 20, 2022.
- T. A. Manz, N. Gabaldon Limas, **T. Chen**, and D. J. Cole, “Accurate methods to describe system-specific polarization and dispersion energies,” 17th Annual URC Research and Creativity Activities Fair, NMSU, Las Cruces, New Mexico, April 4, 2018.
- T. A. Manz, N. Gabaldon Limas, **T. Chen** (co-presenter), and D. J. Cole, “Accurate methods to describe system-specific polarization and dispersion energies,” AIChE Annual Meeting, Minneapolis, Minnesota, October 30, 2017.
- N. Gabaldon Limas, **T. Chen**, T. A. Manz, “Recent Developments in Computing the Properties of Atoms in Materials”, Four Corners + Texas Sections 2016 Meeting of the American Physical Society, Las Cruces, New Mexico, October 21, 2016.

Mentoring Experience

- Prepared instruction documents of computational chemistry software and simulation procedures for new group members.
- Mentored undergraduate students in Dr. Manz’s and Dr. Welborn’s groups on computational chemistry research.
- Teaching assistant of General Chemistry Lab for first year undergraduate students.
- Teaching assistant of Physical Chemistry for junior year chemistry students.
- Teaching assistant of Chemical Separation for junior year chemical engineering students.

Websites

- LinkedIn: www.linkedin.com/in/taoyi-chen-phd-4bb640148
- Google Scholar: scholar.google.com/citations?user=uw7NnEYAAAAJ&hl=en